Closure under the tensor-power reflections makes a tensor set into a Cartan graph

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2023-019-higher-braidings-of-diagonal-type/pinned-original-source.tex (Higher Braidings of Diagonal Type); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author verifies the two Cartan-graph axioms for the reflected set A: (C1) from Proposition RdCartan plus the order-two property of the defining reflection, (C2) by comparing Cartan entries through the R_m factorization and the eigenvector relations of (sigma^q_l)^{tensor dn}, closing via eqref cartan_d. In-span labels: the proposition itself renders as Proposition 3.9; the span carries no numbered display environments. Out-of-span navigation pointers declared as external references: RdCartan (Theorem 3.8, line 734, used at line 756) and cartan_d (Theorem 3.7, line 714, used at line 768 via \eqref). The factorization formula quoted at line 761 is the content of Rossod (3.8), restated inline inside the proof. Seed convention: counters stepped from the source itself and cross-validated against a full host compile of the pinned source (aux ground truth): section 3 of the shared [Theorem] counter holds seven numbered items before this proposition (3.1 through 3.8 with RdCartan = 3.8 and cartan_d = 3.7), and section 3 holds numbered equations (3.1) through (3.8) before the span, so the card prints Proposition 3.9.

Editorial note. The label RdGraph names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own Proposition environment in the article source.

Editorial note. Journal-unit assembly: card-local preamble (author theorem environments and macros verbatim), article class, no bibliography in the counted span (the span carries no \cite commands).

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- Cartan graphs: a quadruple (I, A, (rho_i), (c^q)) with finite index set I, set of objects A, reflections rho_i, and generalized Cartan matrices c^q, satisfying axioms (C1) and (C2) (source Section 3 definitions).

- The tensor-power construction of Section 3: for a tensor q in K^{I^dn} with defined Cartan matrix c^q, the reflection operator sigma^q_l and its eigenvectors v_m, w_m, s_m with the eigenvalue relations (sigma^q_l)^{tensor dn}(v_m) = v_m, (w_m) -> -w_m, (s_m) -> s_m (source Section 3.2, recalled inside the proof at lines 762-764).
- The factorization Rossod (3.8) of the condition R_m = 0 as (1 - bich_q(v_m))(1 - bich_q(w_m))/(1 - chi_q(s_m)), restated inside the proof at line 761, and Theorem cartan_d (3.7) stating that c^q is a generalized Cartan matrix.

Source reference. Original proposition 3.8: sources/sigma-2023-019-higher-braidings-of-diagonal-type/pinned-original-source.tex, source lines 734–734. Navigation pointer invoked at line 756: the counted text cites that numbered proposition of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it; the cited proposition is an allowed input and none of it is reproduced inside the span.

Source reference. Original theorem 3.7: sources/sigma-2023-019-higher-braidings-of-diagonal-type/pinned-original-source.tex, source lines 714–714. Navigation pointer invoked at line 768: the counted text cites that numbered theorem of the same article, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it; the cited theorem is an allowed input and none of it is reproduced inside the span.

Proposition 3.9. *Let $d \in \mathbb{N}$ be even, $I = \{1, \dots, n\}$, and A be a set of tensors $\mathbf{q} \in \mathbb{K}^{I^d}$ with defined Cartan matrix $c^{\mathbf{q}}$. Assume that for all $\mathbf{q} \in A$ and $\ell \in I$ we have*

$$\rho_\ell(\mathbf{q}) := \sigma_\ell^{\mathbf{q}}(\mathbf{q}) \in A.$$

Then $(I, A, (\rho_i)_{i\in I}, (c^{\mathbf{q}})_{\mathbf{q}\in A})$ is a Cartan graph.

Proof. Let $\mathbf{q} \in A$ and $\ell, j \in I$. The matrices $c^{\mathbf{q}}$ are generalized Cartan matrices by Proposition 3.8. If Axiom (C2) is satisfied, then (C1) is satisfied as well because ρ_ℓ is defined using a reflection which has order two and $\sigma_\ell^{\mathbf{q}} = \sigma_\ell^{\rho_\ell(\mathbf{q})}$ since the relevant Cartan entries are equal by (C2).

For (C2), we need to compare $c^{\mathbf{q}}_{\ell,j}$ and $c^{\rho_\ell(\mathbf{q})}_{\ell,j}$ for $\ell \neq j$. With $m := -c^{\mathbf{q}}_{\ell,j}$, we have $R_m = 0$ and $R_k \neq 0$ for $k < m$. Now $R_m = (1-\chi_{\mathbf{q}}(v_m))(1-\chi_{\mathbf{q}}(w_m))/(1-\chi_{\mathbf{q}}(s_m))$. Recall that v_m and w_m are eigenvectors, i.e., $\left(\sigma_\ell^{\mathbf{q}}\right)^{\otimes d}(v_m) = v_m$, $\left(\sigma_\ell^{\mathbf{q}}\right)^{\otimes d}(w_m) = -w_m$; also, $\left(\sigma_\ell^{\mathbf{q}}\right)^{\otimes d}(s_m) = s_m$ because $(m+1)s_m = v_m$. Hence replacing $\mathbf{q}$ by $\sigma_\ell^{\mathbf{q}}(\mathbf{q})$ will map R_m to $(1-\chi_{\mathbf{q}}(v_m))(1-\chi_{\mathbf{q}}(-w_m))/(1-\chi_{\mathbf{q}}(s_m))$. But $1-\chi_{\mathbf{q}}(-w_m) = 0$ if and only if $-w_m = 0$ or equivalently $w_m = 0$. Therefore, via (3.7) the new condition produces the same Cartan entry $c^{\mathbf{q}}_{\ell,j} = c^{\rho_\ell(\mathbf{q})}_{\ell,j}$ for $\ell \neq j$. □